STABILITY PROPERTIES OF MODULES WITH THE FLAT BASE CHANGE PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author marks the tensor-pullback verification in part (5) 'details omitted' with the standard Modules on Sites citation; all other parts are written out.

Allowed standard prerequisites (editorial).

- Restriction comparison on stacks. Sheaves on Stacks, Lemma stacks-sheaves-lemma-comparison. Parts (1), (3), (6).
- Weak Serre characterization and internal hom pullback. Homology, Lemma homology-lemma-characterize-weak-serre-subcategory; Modules on Sites, Lemmas sites-modules-lemma-tensor-product-pullback and sites-modules-lemma-pullback-internal-hom. Parts (2), (5), (6).
- Etale lifting over smooth covers. Sheaves on Stacks, Lemma stacks-sheaves-lemma-surjective-flat-locally-finite-presentation. Part (3).

Source reference. Original reference 075D: sources/stacks-chow/tags.txt, source lines 9127–9127. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0754: sources/stacks-chow/tags.txt, source lines 9118–9118. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 075V: sources/stacks-chow/tags.txt, source lines 9144–9144. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06XH: sources/stacks-chow/tags.txt, source lines 8851–8851. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03EL: sources/stacks-chow/tags.txt, source lines 4460–4460. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WK: sources/stacks-chow/tags.txt, source lines 8819–8819. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0GN2: sources/stacks-chow/tags.txt, source lines 20777–20777. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let $\mathcal{F}$ be an $\mathcal{O}_\mathcal{X}$-module on $\mathcal{X}_{étale}$.*

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(1) *If $\mathcal{F}$ has the flat base change property then for any morphism $g : \mathcal{Y} \to \mathcal{X}$ of algebraic stacks, the pullback $g^*\mathcal{F}$ does too.*
(2) *The full subcategory of $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ consisting of modules with the flat base change property is a weak Serre subcategory.*
(3) *Let $f_i : \mathcal{X}_i \to \mathcal{X}$ be a family of smooth morphisms of algebraic stacks such that $|\mathcal{X}| = \bigcup_i |f_i|(|\mathcal{X}_i|)$. If each $f_i^*\mathcal{F}$ has the flat base change property then so does $\mathcal{F}$.*
(4) *The category of $\mathcal{O}_\mathcal{X}$-modules on $\mathcal{X}_{étale}$ with the flat base change property has colimits and they agree with colimits in $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$.*
(5) *Given $\mathcal{F}$ and $\mathcal{G}$ in $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ with the flat base change property then the tensor product $\mathcal{F} \otimes_{\mathcal{O}_\mathcal{X}} \mathcal{G}$ has the flat base change property.*
(6) *Given $\mathcal{F}$ and $\mathcal{G}$ in $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ with $\mathcal{F}$ of finite presentation and $\mathcal{G}$ having the flat base change property then the sheaf $\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F}, \mathcal{G})$ has the flat base change property.*

Proof. Let $g : \mathcal{Y} \to \mathcal{X}$ be as in (1). Let y be an object of $\mathcal{Y}$ lying over a scheme V. By Sheaves on Stacks, Lemma 075D we have $(g^*\mathcal{F})|_{V_{étale}} = \mathcal{F}|_{V_{étale}}$. Moreover a comparison mapping for the sheaf $g^*\mathcal{F}$ on $\mathcal{Y}$ is a special case of a comparison map for the sheaf $\mathcal{F}$ on $\mathcal{X}$, see Sheaves on Stacks, Lemma 075D. In this way (1) is clear.

Proof of (2). We use the characterization of weak Serre subcategories of Homology, Lemma 0754. Kernels and cokernels of maps between sheaves having the flat base change property also have the flat base change property. This is clear because f_{small}^* is exact for a flat morphism of schemes and since the restriction functors $(-)|_{U_{étale}}$ are exact (because we are working in the étale topology). Finally, if $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ is a short exact sequence of $Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$ and the outer two sheaves have the flat base change property then the middle one does as well, again because of the exactness of f_{small}^* and the restriction functors (and the 5 lemma).

Proof of (3). Let $f_i : \mathcal{X}_i \to \mathcal{X}$ be a jointly surjective family of smooth morphisms of algebraic stacks and assume each $f_i^*\mathcal{F}$ has the flat base change property. By part (1), the definition of an algebraic stack, and the fact that compositions of smooth morphisms are smooth (see Morphisms of Stacks, Lemma 075V) we may assume that each $\mathcal{X}_i$ is representable by a scheme. Let $\varphi : x \to x'$ be a morphism of $\mathcal{X}$ lying over a flat morphism $a : U \to U'$ of schemes. By Sheaves on Stacks, Lemma 06XH there exists a jointly surjective family of étale morphisms $U_i' \to U'$ such that $U_i' \to U' \to \mathcal{X}$ factors through $\mathcal{X}_i$. Thus we obtain commutative diagrams

$$\begin{array}{ccccc} U_i = U \times_{U'} U_i' & \xrightarrow{a_i} & U_i' & \xrightarrow{x_i'} & \mathcal{X}_i \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle f_i} \\ U & \xrightarrow{a} & U' & \xrightarrow{x'} & \mathcal{X} \end{array}$$

Note that each a_i is a flat morphism of schemes as a base change of a. Denote $\psi_i : x_i \to x_i'$ the morphism of $\mathcal{X}_i$ lying over a_i with target x_i'. By assumption the comparison maps $c_{\psi_i} : (a_i)_{small}^*\big(f_i^*\mathcal{F}|_{(U_i')_{étale}}\big) \to f_i^*\mathcal{F}|_{(U_i)_{étale}}$ is an isomorphism. Because the vertical arrows $U_i' \to U'$ and $U_i \to U$ are étale, the sheaves $f_i^*\mathcal{F}|_{(U_i')_{étale}}$ and $f_i^*\mathcal{F}|_{(U_i)_{étale}}$ are the restrictions of $\mathcal{F}|_{U'_{étale}}$ and $\mathcal{F}|_{U_{étale}}$ and the map c_{ψ_i} is the restriction of c_φ to $(U_i)_{étale}$, see Sheaves on Stacks, Lemma 075D. Since $\{U_i \to U\}$ is an étale covering, this implies that the comparison map c_φ is an isomorphism which is what we wanted to prove.

Proof of (4). Let $\mathcal{I} \to Mod(\mathcal{X}_{étale}, \mathcal{O}_\mathcal{X})$, $i \mapsto \mathcal{F}_i$ be a diagram and assume each $\mathcal{F}_i$ has the flat base change property. Let $\varphi : x \to x'$ be a morphism of $\mathcal{X}$ lying over

the flat morphism of schemes $f : U \to U'$. Recall that $\mathrm{colim}_i \mathcal{F}_i$ is the sheafification of the presheaf colimit. As we are using the étale topology, it is clear that

$$(\mathrm{colim}_i \mathcal{F}_i)|_{U_{étale}} = \mathrm{colim}_i \mathcal{F}_i|_{U_{étale}}$$

and similarly for the restriction to $U'_{étale}$. Hence

$$\begin{aligned} f^*_{small}((\mathrm{colim}_i \mathcal{F}_i)|_{U'_{étale}}) &= f^*_{small}(\mathrm{colim}_i \mathcal{F}_i|_{U'_{étale}}) \\ &= \mathrm{colim}_i f^*_{small}(\mathcal{F}_i|_{U'_{étale}}) \\ &\xrightarrow{\mathrm{colim}\, c_\varphi} \mathrm{colim}_i \mathcal{F}_i|_{U_{étale}} \\ &= (\mathrm{colim}_i \mathcal{F}_i)|_{U_{étale}} \end{aligned}$$

For the second equality we used that f^*_{small} commutes with colimits (as a left adjoint). The arrow is an isomorphism as each $\mathcal{F}_i$ has the flat base change property. Thus the colimit has the flat base change property and (4) is true.

Part (5) holds because tensor products commute with pullbacks, see Modules on Sites, Lemma 03EL. Details omitted.

Let $\mathcal{F}$ and $\mathcal{G}$ be as in (6). Since $\mathcal{F}$ is quasi-coherent it has the flat base change property by Sheaves on Stacks, Lemma 06WK. Let $\varphi : x \to x'$ be a morphism of $\mathcal{X}$ lying over the flat morphism of schemes $f : U \to U'$. As we are using the étale topology, we have

$$\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F}, \mathcal{G})|_{U_{étale}} = \mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_{U_{étale}}, \mathcal{G}|_{U_{étale}})$$

and similarly for the restriction to $U'_{étale}$ (details omitted). Hence

$$\begin{aligned} f^*_{small}(\mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F}, \mathcal{G})|_{U'_{étale}}) &= f^*_{small}(\mathcal{H}om_{\mathcal{O}_{U'}}(\mathcal{F}|_{U'_{étale}}, \mathcal{G}|_{U'_{étale}})) \\ &= \mathcal{H}om_{\mathcal{O}_{U'}}(f^*_{small}(\mathcal{F}|_{U'_{étale}}), f^*_{small}(\mathcal{G}|_{U'_{étale}})) \\ &\xrightarrow{c_\varphi} \mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_{U_{étale}}, \mathcal{G}|_{U_{étale}}) \\ &= \mathcal{H}om_{\mathcal{O}_\mathcal{X}}(\mathcal{F}, \mathcal{G})|_{U_{étale}} \end{aligned}$$

Here the second equality is Modules on Sites, Lemma 0GN2 which uses that $f : U \to U'$ is flat and hence the morphism of ringed sites f_{small} is flat too. The arrow is an isomorphism as both $\mathcal{F}$ and $\mathcal{G}$ have the flat base change property. Thus our $\mathcal{H}om$ has the flat base change property too as desired. □